Pro Football Sports Science Research Update

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'Post-rehabilitation phase' in professional football: Are we optimizing player support after return to play? British Journal of Sports Medicine, Ahead of Print: 1-3, 2025.

A review of syndesmosis injuries and preferred treatment in football players. Current Reviews in Musculoskeletal Medicine, Ahead of Print: 1-11, 2025.

Between-session reliability of field-based assessments of isometric neck strength. Sensors, 24: 5015, 2024.

Blood flow restriction: Methods and apparatus still matter. British Journal of Sports Medicine, Ahead of Print: 1-3, 2025.

Characterization of distal biceps tendon and triceps tendon injuries in National Football League players from 2009 to 2022. Journal of Orthopaedics, 68: 20-26, 2025.

Classification of lower body strength qualities: A data-driven approach. International Journal of Sports Science & Coaching, Ahead of Print: 1-24, 2025.

Clinical considerations for competitive sports participation for athletes with cardiovascular abnormalities. Journal of the American College of Cardiology, Ahead of Print: 1-50, 2025.

Clinically feasible dual-task detects residual post-concussion deficits after return-to-play. Ahead of Print: 1-9, 2024.

Confirmation bias in sport science: Understanding and mitigating its impact. International Journal of Sports Physiology and Performance, Ahead of Print: 1-6, 2025.

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Cross-validation of the safe supplement screener (S3) predicting consistent third-party-tested nutritional supplement use in NCAA Division 1 athletes. Frontiers in Nutrition, 11:1519544, 2025.

Does accumulated physical load in different time windows affect hamstring injuries in elite football players? Research in Sports Medicine, Ahead of Print: 1-14, 2025.

Dry needling in sports and sport recovery: A systematic review with an evidence gap map. Sports Medicine, Ahead of Print: 1-34, 2025.

Effect of sleep duration and concussion history on neurocognitive testing symptoms for post-injury states. Clinical Journal of Sport Medicine, Ahead of Print: 1-7, 2024.

Evening alpha-lactalbumin supplementation alters sleep architecture and reduces morning reaction time in an athletically trained population with sleep difficulties. International Journal of Sport Nutrition and Exercise Metabolism, Ahead of Print: 1-10, 2025.

Glutamine supplementation: A possible strategy to help mitigate health risks of heat-related illness. Journal of Nutrition and Metabolism, Article 1638244, 2024.

Head in the game: The impact of cognitive abilities on performance of National Football League quarterbacks. Frontiers in Psychology, 16:1540498, 2025.

Hip adduction and abduction strength values in NCAA Division 1 American football players: Strength values and a comparison across position groups. Physical Therapy in Sport, Ahead of Print: 1-30, 2025.

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Recent trends in quadriceps tendon and patellar tendon injuries in the National Football League. The Orthopaedic Journal of Sports Medicine, 13(2): 1-9, 2025.

Relationships between the pain intensity at the time of the hamstring muscle injury and the hyperintense T2 weighted volume of the hamstring muscle lesion measured by MRI. Clinical Journal of Sport Medicine, Ahead of Print: 1-6, 2024.

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'Post-rehabilitation phase' in professional football: Are we optimizing player support after return to play? British Journal of Sports Medicine, Ahead of Print: 1-3, 2025.

Overview:

1. Return to Play (RTP) Decisions: The decision to allow a player to return to play is a balance between risk and reward. Early RTP can enhance team performance but also increases the risk of subsequent injuries.
2. Subsequent Injury: Previous injuries significantly increase the risk of future injuries. Subsequent injuries are a poorly researched area despite their high occurrence rates in professional football. Inadequate rehabilitation or premature RTP often lead to such injuries.
3. RTP Frameworks: Modern RTP frameworks emphasize a competency-based, evolving process rather than a single decision point. The frameworks include a multi-stage process from diagnosis to full RTP.
4. Post-Rehabilitation Phase: A proposed addition to existing RTP frameworks, this phase would run concurrently with final rehabilitation stages and focus on specific monitoring to mitigate subsequent injury risks. Tools like GPS data and neuromuscular diagnostics are suggested for better monitoring.
5. Future Directions: There is a need for more research to understand subsequent injury risks and develop specific monitoring and loading strategies. This research can help inform better rehabilitation practices across various sports.

This editorial underscores the importance of ongoing rehabilitation and monitoring to minimize injury risks after players return to play.

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A review of syndesmosis injuries and preferred treatment in football players. Current Reviews in Musculoskeletal Medicine, Ahead of Print: 1-11, 2025.

Purpose of review: Syndesmotic injuries are common football injuries. The unique demands of football athletes create large magnitude rotational moments about the ankle, even during low impact maneuvers. This review explores the structure and function of the syndesmosis, assesses recent data in football athletes at the professional and collegiate levels regarding epidemiology, describes available treatment options, and provides example cases from the authors' institution. The review concludes with clinical and surgical pearls for the evaluation and treatment of syndesmotic injury.

Recent findings: In general, flexible syndesmotic fixation has demonstrated similar clinical outcome scores as rigid fixation. Flexible fixation has demonstrated benefit over rigid fixation in terms of implant failure, hardware removal, and local irritation. Both flexible and rigid fixation remain viable options for treatment of syndesmotic injuries yet the indications for selecting a construct are often subjective. Certain cases of high-risk football players such as linemen may warrant careful consideration of rigid fixation options despite the clinical advantages of flexible fixation. During fixation, direct visualization techniques with open or arthroscopic assistance for reduction of the syndesmosis remain superior and enable diagnosis of chondral defects. Flexible and rigid syndesmotic fixation techniques are viable for treatment of unstable syndesmotic injuries in athletes. Recent literature favors flexible fixation. However, at-risk football athletes or those with length unstable fibula fractures may benefit from rigid or supplemental flexible fixation as opposed to traditional flexible fixation. We recommend direct visualization of reduction at the syndesmosis during surgical treatment of unstable ankle injuries.

Between-session reliability of field-based assessments of isometric neck strength. Sensors, 24: 5015, 2024.

Abstract - The reliability of the fixed-frame dynamometer for measuring isometric neck strength is established, but with limited field-based applications. This study aimed to establish the inter-and intra-session reliability of the peak force for neck flexors, extensors, and side flexors using the VALD ForceFrame and DynaMo and the force–time characteristics in the quadruped position (ForceFrame). Twenty-seven recreationally active males performed three repetitions of isometric neck flexion,

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extension, and side flexion over two sessions in random order using the VALD ForceFrame and DynaMo. Both devices demonstrated acceptable reliability, with the Forceframe ICC > 0.8 and CV% < 13.8% and the DynaMo ICC > 0.76 and CV% < 13.8%. No systematic or proportional differences were found using the Passing–Bablock procedure, and Bland–Altman analysis confirmed agreement across measures. Reliability was shown for right-side (ICC > 0.76) and left-side (ICC > 0.79) flexion and flexion (ICC > 0.75) across 50, 100, 150, and 200 ms. Statistical parametric mapping indicated no differences in ForceFrame-generated isometric force–time curves between sessions, though the CV was highest in the force development phase. The findings suggest that both tools can reliably assess neck strength, supporting their use in sports and clinical settings. However, assessment methods are not interchangeable, emphasizing the need for standardized neck strength assessment approaches.

Blood flow restriction: Methods and apparatus still matter. British Journal of Sports Medicine, Ahead of Print: 1-3, 2025.

1. **Overview of Blood Flow Restriction (BFR)**: BFR is used to enhance health, performance, rehabilitation, and recovery. It involves using a pneumatic tourniquet cuff on the limbs to partially restrict arterial inflow and completely restrict venous outflow, creating a hypoxic environment that stimulates adaptive responses. Non-pneumatic methods also exist, but they pose challenges in ensuring reliable and safe pressure application.
2. **Impact on Psychophysiological Responses**: Key methodological variables, such as cuff width and pressure, impact the BFR stimulus and subsequent psychophysiological responses. Guidelines have been developed to optimize BFR application, recommending wider cuffs and pressure relative to the individual's limb occlusion pressure to minimize discomfort and maximize safety and efficacy.
3. **Commercial BFR Systems**: The growing interest in BFR has led to an increase in commercially available systems with new features like autoregulation capabilities and different types of bladder systems. These features can impact the BFR stimulus and acute psychophysiological responses.

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4. **Reporting Recommendations**: There is a need for thorough and transparent reporting of BFR apparatus and methodology in scientific literature to ensure consistency, safety, and efficacy. The document provides specific reporting recommendations for researchers to consider in their studies.

The authors emphasize the importance of detailed reporting on BFR apparatus and methodology to advance research and practice in this field.

Characterization of distal biceps tendon and triceps tendon injuries in National Football League players from 2009 to 2022. Journal of Orthopaedics, 68: 20-26, 2025.

Purpose: The purpose of this study is to determine the incidence and impact of distal biceps and triceps tendon injuries in the National Football League (NFL) from the 2009-10 to the 2022-23 seasons. This study explores the impact that player and injury characteristics have on injury risk, return-to-play, and player performance.

Methods: Data from the 2009-10 to the 2022-23 NFL seasons were analyzed for players with distal biceps or triceps tendon tears. Return-to-play (RTP) and performance metrics were recorded for each player during the season before and first two post-injury seasons. Data were analyzed to determine statistically significant differences in proportions using chi-square, Fisher's exact, or McNemar tests. Statistical significance was set at $p < 0.05$.

Results: Fifty tendon ruptures (26 biceps, 24 triceps) were identified. Biceps injuries were more common in defensive players (73.1 %), while triceps injuries predominantly affected offensive players (58.3 %). Significant risk factors for biceps injuries included BMI ≥31 ($p = 0.0008$) and ≥4 seasons of experience ($p = 0.031$, OR = 2.7). Triceps injuries were associated with BMI ≥31 ($p = 0.01$), age ≥26 ($p < 0.0001$), and ≥4 seasons of experience ($p = 0.006$). RTP rates were 73.1 % for biceps and 70.8 % for triceps injuries. However, only 52.6 % and 41.2 % of players with biceps and triceps injuries, respectively, returned to pre-injury performance levels. Younger players (<26 years) and those with fewer years of experience (≤4 years) were more likely to achieve prior performance levels.

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Conclusion: Distal biceps and triceps tendon injuries are typically season-ending for NFL players. Significant risk factors include BMI ≥31, ≥4 years of NFL experience, and game exposure, with age ≥26 being an additional risk factor for triceps tendon injuries. RTP rates exceed 70 %, indicating a strong potential for athletes to return to the NFL post-injury.

Classification of lower body strength qualities: A data-driven approach. International Journal of Sports Science & Coaching, Ahead of Print: 1-24, 2025.

Abstract - Maximal strength of the lower body is important for many athletic tasks and demands. Lower body extensor strength, defined as maximal expressions of force in the vertical plane, can be measured in several different ways, using a wide range of tests and metrics. Although there have been investigations into the interrelationships between lower body strength and force measures, there is no known analysis or synthesis of these data that focuses on classifying variables into groups based on statistical similarity and/or distinction. This information is important to consider in an environment where many measurement options exist for diagnosing lower body characteristics. Understanding the overlap and distinction of these analysis methods and metrics will inform parsimonious, yet comprehensive, assessment of the lower body. It was therefore the purpose of this review to explore and summarize the commonality (i.e., r, R^2, component loading) between metrics from weightbearing, multi-joint lower body strength tests that are presented in the current literature, with the aim to group metrics into distinct domains of strength expression. The results of this review suggest that five unique domains of lower body extensor strength are present in athletes and trained populations: 1) reactive dynamic, 2) unloaded dynamic, 3) loaded dynamic, 4) early isometric, and 5) maximal isometric. Despite these findings, there are considerable limitations and gaps in the literature, and it is suggested that additional empirical investigations be conducted to better understand this concept of lower body strength domains.

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Clinical considerations for competitive sports participation for athletes with cardiovascular abnormalities. Journal of the American College of Cardiology, Ahead of Print: 1-50, 2025.

Abstract - This American Heart Association/American College of Cardiology scientific statement on clinical considerations for competitive sports participation for athletes with cardiovascular abnormalities or diseases is organized into 11 distinct sections focused on sports-specific topics or disease processes that are relevant when considering the potential risks of adverse cardiovascular events, including sudden cardiac arrest, during competitive sports participation. Task forces comprising international experts in sports cardiology and the respective topics covered were assigned to each section and prepared specific clinical considerations tables for practitioners to reference. Comprehensive literature review and an emphasis on shared decision-making were integral in the writing of all clinical considerations presented.

Clinically feasible dual-task detects residual post-concussion deficits after return-to-play. Ahead of Print: 1-9, 2024.

Abstract - Dual-task (DT) has been shown to detect post-concussion deficits even after traditional measures returned to normal. However, previous studies were conducted in laboratory settings that were not feasible in the clinical setting. Prospective Cohort Study. Two groups (concussion and control groups) completed two testing sessions. The concussion group was tested at 7–10 days post-concussion (subacute) and after Return-to-Play (RTP). The control group was tested with similar time windows. Participants completed 2 single task (ST): Expanded Timed Get Up and Go (ETGUG) as a physical task and Auditory Pure Switch Task (APST) as a cognitive task. After completing STs, participants completed DT performing ETGUG and APST simultaneously. A mixed method ANOVA was conducted to analyze the effects of concussion on ST and DT performance. Another mixed method ANOVA was conducted to examine the effect of concussion recovery time. The concussion group took significantly longer to complete ST and DT than controls at the subacute phase ($p = 0.018$, $p = 0.014$, respectively). At RTP timepoint, the concussion group took significantly longer to complete ST and DT than the controls

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(p=0.007, p < 0.001, respectively). The deficits were more clearly shown for DT. Clinically feasible DT detected subtle and residual post-concussion deficits after athletes were cleared to RTP.

Confirmation bias in sport science: Understanding and mitigating its impact. International Journal of Sports Physiology and Performance, Ahead of Print: 1-6, 2025.

Purpose: Confirmation bias, a cognitive phenomenon deeply ingrained in human thinking, shapes our perception of reality. It leads individuals to favor information that aligns with their preexisting beliefs while dismissing or undervaluing the contrary evidence. In sport science, where precision and evidence-based practices are critical aspects to optimize performance, confirmation bias can decisively impact decision-making processes. In this context, practitioners rely on both objective and subjective data to be interpreted to enhance athletic performance. However, confirmation bias introduces a subtle yet powerful distortion, which is difficult to control and quantify. We usually believe that our observations can justify expectations or predictions about future observations, as well as broader generalizations beyond what we have directly observed—inductive inferences. Researchers and practitioners may therefore unknowingly seek out evidence that confirms their beliefs, inadvertently overlooking alternative viewpoints and interpretations.

Conclusions: This paper explores the concept of confirmation bias in sport science, providing practical examples in the contexts of testing, training, and monitoring. By understanding how confirmation bias operates (ie, heuristic of decision making), we may develop strategies to mitigate its effects and promote more accurate and effective decision making in sport science (eg, performance).

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Correlation of King-Devick test and helmet impact exposures over a youth football season. Sports Health, Ahead of Print: 1-9, 2025.

Background: The cumulative effect of repetitive subconcussive head impacts on neurocognitive function during youth contact sports remains largely unknown. There is a paucity of literature evaluating cumulative helmet forces over a season and their correlation with preseason and postseason cognitive performance tasks such as the King-Devick test (KDT).

Hypothesis: Higher helmet forces recorded throughout a 10-week, 10-game youth football season would correlate with slower performance on postseason KDT.

Study design: Prospective cohort study.

Level of evidence: Level 3.

Methods: A cohort of 58 youth football players (ages 9-13 years) underwent pre- and postseason KDT. Players wore SpeedFlex helmets (Riddell) fitted with InSite Impact Response System helmet accelerometers (Riddell) which recorded impacts of ≥15g. Head impacts were tallied over a season and assigned a score of 1, 2, or 3 based on magnitude of g forces. Suspected concussions were correlated with KDT times and recorded instances of head impact. Pre- and postseason KDT scores were compared.

Results: During the season, 2013 head impacts were recorded. Median (range) total cumulative force score was 24 (5-476); 6 players sustained head impacts concerning for concussion, and 4 were clinically diagnosed with concussions. Overall, postseason KDT times improved compared with preseason, with a median (range) change of -4.8 seconds (-7.6, -1.1). Analysis showed no correlation between changes in KDT time and total cumulative force score over the season.

Conclusion: KDT times in youth football players did not change significantly based on head impact exposure over a single youth football season; most players' KDT times improved from preseason to postseason.

Clinical relevance: Although our study did not show significant cognitive impact as measured by KDT over a single youth football season, the long-term effects of concussion on the immature brain and how it can impact cognitive development remains largely unknown and should be an area of ongoing study.

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Cross-validation of the safe supplement screener (S3) predicting consistent third-party-tested nutritional supplement use in NCAA Division 1 athletes. Frontiers in Nutrition, 11:1519544, 2025.

Introduction: This cross-sectional study aimed to cross-validate an earlier developed algorithm-based screener and explore additional potential predictors for whether athletes will use third-party-tested (TPT) supplements.

Methods: To justify the initial model behind the supplement safety screener (S3) algorithm which predicts whether athletes will use TPT supplements, a cross-validation was performed using this independent dataset based on responses of a large group of collegiate NCAA DI athletes. Additionally, explorative modeling using stepwise logistic regression was used to identify new predictors for TPT supplement use to create and evaluate a new model for future use.

Results: The S3 algorithm was applied to the responses from n = 662 athletes using supplements (age: 20 ± 1.5 years, 50% female, from >24 sports) confirming that the algorithm identifies consistent and inconsistent TPT users ($\chi2$ (1) = 15.95, $p < 0.001$), with a moderate area under the curve (AUC, 0.67) and a moderate specificity (68%), but low sensitivity (51%). Explorative modeling identified two new variables: TPT logo recognition, and having at least one name, image, likeness (NIL) deal that may help to predict TPT supplement use.

Discussion: Strong relationships between risk groupings and product use outcomes toward TPT supplement use were identified for the athletes screened. The S3 screener showed high sensitivity for identifying student-athletes inconsistently using TPT supplements, but a low specificity, lacking the ability to place less risky athletes into low-risk quadrants. The exploratory modeling, identifying TPT logo recognition and having a NIL deal, further strengthens our knowledge on predictors for consistent TPT supplement use.

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Does accumulated physical load in different time windows affect hamstring injuries in elite football players? Research in Sports Medicine, Ahead of Print: 1-14, 2025.

Background: This study aims to investigate how accumulated training load over different time windows (7, 14, and 28 days) influences the incidence of hamstring injuries in elite football players.

Methods: A descriptive and longitudinal study was developed in 343 players in three teams (2nd Team, U-19 team, and U-18 team) from the academy of an elite football club in Spanish La Liga that also competed in the Union of European Football Associations Champions League during four seasons, from 2017/18 to 2020/21. External load variables (total distance, high-speed running, accelerations, etc.) for training and competition have been studied considering the volume variables during three different time windows along the different seasons (7 days, 14 days, and 28 days).

Results: Results did not show differences in the shortest time windows (7- and 14-week periods), showing that strength and conditioning coaches should analyze longer periods of time and compare them to previous periods with the same length in order to improve workload management and reduce injury risk.

Conclusions: It is important consider the accumulated load (chronic) in periods of 28 days to avoid an increase that generates a hamstring injury. The control of the training load in blocks of 28- day time windows seem essential to minimize the injury rate in football players.

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Dry needling in sports and sport recovery: A systematic review with an evidence gap map. Sports Medicine, Ahead of Print: 1-34, 2025.

Background: Dry needling is an increasingly popular technique used in sports and regenerative medicine contexts. However, there is no comprehensive overview of investigations of dry needling in sports and sport recovery.

Objectives: The objectives were to perform a systematic review of dry needling in sports athletes with an evidence gap map, to identify current gaps in the literature, and to provide stakeholders with direction for future research.

Methods: Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) 2020 guidelines were followed. Studies on healthy and injured athletes receiving dry needling were included. Three databases (PubMed, Scopus, and Web of Science) were searched, data were synthesized narratively, key data were summarized quantitatively, and an evidence gap map was created.

Results: The authors incorporated 24 studies into the current study, encompassing 580 athletes, predominantly involving talent/developmental and highly trained/national-level athletes across 13 sports. Most studies used passive or placebo controls, with 37.5% incorporating active controls. Interventions focused mainly on the lower limbs (58.3%). Around 69% of studies reported pain perception outcomes, while six examined muscular strength, activity, and range of motion. While results varied, dry needling generally showed a more positive effect on pain than on athletic performance.

Conclusions: Dry needling studies seem to describe general effectiveness and safety for reducing pain and muscle stiffness in a wide variety of body regions. However, further research is needed on underrepresented populations such as elite, world-class, and Paralympic athletes, as well as expanding investigations into long-term effects and a broader range of muscle groups, particularly the hamstrings. The results may be valuable for medical professionals, sports medicine specialists, and researchers.

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Effect of sleep duration and concussion history on neurocognitive testing symptoms for post-injury states. Clinical Journal of Sport Medicine, Ahead of Print: 1-7, 2024.

Objective: Evaluate the influence of sleep duration and concussion history on postconcussion symptoms in adolescent athletes.

Design: Observational retrospective study using the Immediate Post-Concussion Assessment and Cognitive Testing and Post-Concussion Symptom Scale (PCSS). Multivariable linear regression assessed sleep hours against 22 PCSS symptoms, controlling for demographic and health variables.

Setting: Urban concussion centers in Colorado and Florida, 2009 to 2019.

Participants: 11 564 student-athletes aged 12 to 22, categorized by concussion history.

Interventions: Analysis of sleep duration and concussion history in relation to neurocognitive and psychiatric symptom severity.

Main outcome measures: Primary outcomes included neurocognitive, psychiatric, and total symptom scores. Secondary outcomes were specific PCSS symptoms.

Results: Among 5349 student-athletes, 2671 (49.9%) had no prior concussions and 2678 (50.1%) had 1 or more. For those without prior concussions, sleep was negatively associated with age ($\beta = -0.18$, 95% confidence interval [CI], -0.22 to -0.13, $P < 0.0001$), vomiting ($\beta = -0.22$, 95% CI, -0.38 to -0.05, $P = 0.012$), and difficulty concentrating ($\beta = -0.11$, 95% CI, -0.19 to -0.35, $P = 0.005$). In athletes with a history of concussion, less sleep correlated with decreased age ($\beta = -0.11$, 95% CI, -0.14 to -0.07, $P < 0.0001$), headache ($\beta = -0.065$, 95% CI, -0.12 to -0.01, $P = 0.031$), irritability ($\beta = -0.08$, 95% CI, -0.15 to -0.01, $P = 0.021$), and difficulty concentrating ($\beta = -0.08$, 95% CI, -0.15 to -0.01, $P = 0.031$) but increased sensitivity to light ($\beta = 0.10$, 95% CI, 0.001-0.137, $P = 0.048$), numbness/tingling ($\beta = 0.15$, 95% CI, 0.04-0.26, $P = 0.008$), and feeling slowed down ($\beta = 0.13$, 95% CI, 0.05-0.21, $P = 0.001$).

Conclusion: Sleep duration and concussion history are associated with variations in postconcussion symptom severity among adolescent athletes. This data underscores the need for individualized management strategies based on sleep patterns and concussion history.

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Evening alpha-lactalbumin supplementation alters sleep architecture and reduces morning reaction time in an athletically trained population with sleep difficulties. International Journal of Sport Nutrition and Exercise Metabolism, Ahead of Print: 1-10, 2025.

Abstract - Evening consumption of a whey protein rich in the amino acid tryptophan, alpha-lactalbumin (ALAC), has previously shown to benefit sleep—particularly among poor sleepers. Given trained populations often experience sleep difficulty, this study investigated whether evening supplementation of ALAC would influence sleep outcomes, mood, and next-day cognitive performance within a trained population with sleep difficulties. Nineteen trained participants (females, n = 11) with sleep difficulties (Athlete Sleep Screening Questionnaire: 8.1 ± 3.1; Pittsburgh Sleep Quality Index: 10.5 ± 4.1) completed this double-blinded, counterbalanced, randomized, crossover trial. Forty grams of ALAC or control were supplemented 2 hr pre-sleep for three consecutive nights in a controlled environment, with sleep measured using dry electroencephalography. Blood samples were taken on the first evening of each experimental trial, with mood, sleepiness, and recovery assessed across the evening and morning. A cognitive testing battery was also completed each morning. During the ALAC condition, the primary findings were that participants had raised plasma tryptophan levels ($p < .01$), increased nonrapid eye movement Stage 2 sleep duration (CON: 205.9 ± 33.3; ALAC: 216.5 ± 33.1 min), reduced rapid eye movement duration (CON: 110.8 ± 27.9; ALAC: 99.7 ± 23.1 min), and improved reaction time in cognitive tests involving sensory motor speed, spatial orientation, and vigilant attention ($p < .05$). Data suggest evening supplementation of 40 g ALAC alters sleep architecture and improves next-morning reaction time in trained populations with sleep difficulties. Therefore, trained individuals experiencing sleep difficulty may benefit from acute ALAC supplementation to assist next-day performance. Future research should investigate this effect within habitual environments, outside of a tightly controlled setting.

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Glutamine supplementation: A possible strategy to help mitigate health risks of heat-related illness. Journal of Nutrition and Metabolism, Article 1638244, 2024.

Abstract - A rise in body temperature caused by physical work, including exercise, in a hot climate can lead to heat-related illnesses such as exertional heat exhaustion and stroke. Individuals who work physically demanding occupations in hot environments are at heightened risk of heat injury. The mechanisms that contribute to heat illness resulting from physical work in the heat are complex and include dehydration, tissue ischemia and damage, oxidative stress, and inflammatory events. Therefore, it is important to develop strategies that address these mechanistic underpinnings to prevent exacerbation to heat illness. Glutamine is an amino acid that has been considered conditionally essential during situations of biological stress (e.g., tissue burn, exercise, sepsis) due to high rates of tissue consumption. Evidence suggests that glutamine may serve as an important nutrient during heat stress and when combined with other preventative measures (e.g., cooling techniques, work/rest ratios, clothing) may help to mitigate heat illness among individuals working in extreme climates. The aim of this review is to examine the current literature on the role of glutamine during heat stress.

Head in the game: The impact of cognitive abilities on performance of National Football League quarterbacks. Frontiers in Psychology, 16:1540498, 2025.

Abstract - American football is a multi-billion-dollar industry and source of social identity and national pride. Recruiting top level players is a priority for franchises, coaches, teams, and fans. Utilizing data obtained from 42 National League Football (NFL) quarterbacks, collected at their respective Combine experience, the current study adds to existing research demonstrating that cognitive abilities, as measured by the Athletic Intelligence Quotient (AIQ), namely Visual Spatial Processing, Reaction Time, and Decision Making, all increase the predictive accuracy beyond the role of draft pick at the Combine. Reaction Time; Visual Spatial Processing and Decision Making to a lesser, but notable degree; predicted NFL performance metrics such as Career Approximate Value, Quarterback Rating, passing and rushing yards per game, turnover worthy plays, and throwing accuracy. The role of cognitive abilities, particularly in the critical position of quarterback in American football, is discussed.

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Hip adduction and abduction strength values in NCAA Division 1 American football players: Strength values and a comparison across position groups. Physical Therapy in Sport, Ahead of Print: 1-30, 2025.

Objectives: To report hip adduction, abduction, and adduction:abduction strength ratio values in NCAA Division I American football players, and to compare strength values across position groups.

Design: Cross-sectional study.

Setting: University training facility.

Participants: 85 male football players.

Main outcome measures: Isometric hip adduction and abduction strength values in the 0° hip-flexion long-lever and 0° hip-flexion short-lever testing positions, using a ForceFrame.

Results: Hip strength values are presented across two testing positions. ANOVA revealed significant differences ($p \leq 0.05$) in hip strength between position groups, with medium-to-large effect size ($n2 = 0.116$-0.284). Skill-position and mid-position players demonstrated significantly greater hip adduction and abduction relative strength in both testing positions, when compared to lineman-position players. In addition, skill-position and mid-position players demonstrated significantly greater hip adduction:abduction strength ratios (0.19-0.20, 24-25%) in the long-lever testing position, when compared to lineman-position players, with large effect size ($n2 = 0.178$). No significant differences in strength were observed between skill-position and mid-position players. Hip adduction:abduction strength ratios ranged from 0.68 to 0.88 in the long-lever testing position and 1.05 to 1.09 in the short-lever testing position.

Conclusion: Significant differences in relative strength exist between position groups of NCAA Division-I football players, in hip adduction, abduction and adduction:abduction ratio.

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Injury differences between youth male and female American football players. The Physician and Sportsmedicine, Ahead of Print: 1-18, 2025.

Objectives: American football continues to be a popular and rapidly growing sport among both males and females, with relatively little research concerning injuries for the latter – especially in the youth population. The current study aims to evaluate the differences in the epidemiology of injured body parts and injury diagnoses between youth male and female American football players.

Methods: Injury data for male and female American football players were collected over a 10- year period from the National Electronic Injury Surveillance System (NEISS) and retrospectively reviewed. Patients between ages 4 and 18 years with a medical injury/condition related to American football were included. Descriptive analyses were performed for all data including injured body parts, injury diagnoses and setting, demographics, and medical disposition. Odds ratio (OR), 95% CI, and p-values ($p < 0.05$) were used for the statistical significance between the two groups.

Results: A total of 82,174 American football injuries were found in the NEISS dataset, 95.0% of which were male with a mean age of 13.0 ± 2.9 years. The most injured body parts were the head (17.7%), finger (13.8%), knee (9.2%), shoulder (8.7%), and ankle (8.0%). The most prevalent injury diagnoses were fractures (24.5%), sprains/strains (22.7%), contusions/bruises (12.2%), concussions (9.8%), and internal injuries (7.1%). Males were more likely to suffer head (OR = 1.32, $p < 0.001$), knee (OR = 1.23, $p < 0.001$) and shoulder (OR = 2.19, $p < 0.001$) injuries and females suffered more finger (OR = 3.22, $p < 0.001$) and ankle (OR = 1.25, $p < 0.001$) injuries. Males were more likely to suffer concussions (OR = 1.50, $p < 0.001$) whereas females sustained more sprains/strains (OR = 1.32, $p < 0.001$).

Conclusion: Youth male American football players had a greater likelihood of sustaining concussions and head, knee, and shoulder injuries, whereas females were more likely to have sprains/strains, finger, and ankle injuries. Future studies are warranted to further explore how these differences may be attributed to play style and sex-based physiologic and development differences.

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Microbial Champions: The influence of gut microbiota on athletic performance via the gut-brain axis. Open Access Journal of Sports Medicine, 15: 209-228, 2024.

Abstract - In recent years, exercise has shown a powerful ability to regulate the gut microbiota received with concern. For instance, compared with the sedentary group, high-level athletes showed a different gut microbiota composition and remarkable capability of physiological metabolism. In addition, different diet patterns (eg, high-fat diet, high carbohydrate diet et.al) have different effects on gut microbiota, which can also affect exercise performance. Furthermore, adaptations to exercise also might be influenced by the gut microbiota, due to its important role in the transformation and expenditure of energy obtained from the diet. Therefore, appropriate dietary supplementation is important during exercise. And exploring the mechanisms by which dietary supplements affect exercise performance by modulating gut microbiota is of considerable interest to athletes wishing to achieve health and athletic performance. In this narrative review, the relationship between gut microbiota, dietary supplements, training adaptations and performance is discussed as follows. (i) The effects of the three main nutritional supplements on gut microbiota and athlete fitness. (ii) Strategies for dietary supplements and how they exerted function through gut microbiota alteration based on the gut-brain axis. (iii) Why dietary supplement interventions on gut microbiota should be tailored to different types of exercise. Our work integrates these factors to elucidate how specific nutritional supplements can modulate gut microbiota composition and, consequently, influence training adaptations and performance outcomes, unlike previous literature that often focuses solely on the effects of exercise or diet independently. And provides a comprehensive framework for athletes seeking to optimize their health and performance through a microbiota-centric approach.

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National Football League Scouting Combine test results of elite-level National Football League players. Research Quarterly for Exercise and Sport, Ahead of Print: 1-11, 2025.

Abstract - This study aimed to determine elite and poor performance for each offensive and defensive position group on every physical test performed at the NFL Scouting Combine Descriptive statistics were calculated as well as independent samples t-tests for the 15 position groups relative to their performance in the NFL Scouting Combine tests, including: (a) 40-yard Dash, (b) Vertical Jump, (c) Bench Press, (d) Broad Jump, (e) Three-cone Drill, and (f) Shuttle Run. Each player's height and weight were also analyzed. The 15 position groups include Quarterbacks (n = 206), Fullbacks (n = 64), Running Backs (n = 314), Wide Receivers (n = 501), Tight Ends (n = 187), Offensive Tackles (n = 279), Guards (n = 198), Centers (n = 99), Cornerbacks (n = 374), Free Safeties (n = 132), Strong Safeties (n = 129), Outside Linebackers (n = 273), Inside Linebackers (n = 162), Defensive Ends (n = 280), and Defensive Tackles (n = 273). For each of the 15 position groups, there was at least one NFL Scouting Combine test result that had a statistically significant difference at the $p < .05$ or $p < .001$ level when comparing players who averaged playing in at least 10 games per season versus those that played in less than 10 games. The results of this study provide knowledge of which physical test is essential for each offensive and defensive position and can be helpful for those charged with making player personnel decisions in NFL organizations.

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Neurocognitive and Neuromuscular Rehabilitation Techniques after ACL Injury, Part 1: Optimizing Recovery in the Acute Post-Operative Phase- A Clinical Commentary. International Journal of Sports Physical Therapy, 19(11): 1373-1385, 2024.

Abstract - Anterior cruciate ligament (ACL) injury rates are on the rise, despite improved surgical techniques and prevention programs. While traditional rehabilitation emphasizes the restoration of motion, strength, and physical performance, emerging research highlights the importance of addressing neurocognitive deficits that can persist after injury. These deficits, including altered proprioception, impaired motor control and muscle recruitment, as well as heightened reliance on visual feedback, can significantly increase the risk of re-injury and impede return to sport. The purpose of this clinical commentary is to outline a proposed comprehensive approach to rehabilitation that challenges the neurocognitive system to optimize rehabilitation outcomes and reduce reinjury risk. Thus, this clinical commentary discusses the rationale for integrating neurocognitive training into all phases of ACLR rehabilitation, from initial injury to eight weeks post-surgery. It details the neurophysiological changes caused by ACL injury and presents evidence supporting the use of exercises that challenge visual attention, decision-making, and motor planning. A comprehensive rehabilitation framework incorporating both physical and neurocognitive components is proposed, aiming to improve long-term outcomes and reduce re-injury risk.

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Neurocognitive and Neuromuscular Rehabilitation Techniques after ACL Injury – Part 2: Maximizing Performance in the Advanced Return to Sport Phase. International Journal of Sports Physical Therapy, 19(12): 1629-1640, 2024.

Background: Anterior cruciate ligament (ACL) injury and reinjury rates are on the rise, despite improved surgical techniques and prevention programs. ACL injuries also lead to a variety of neuroplastic and neuromuscular alterations. Emerging research highlights the importance of addressing neurocognitive deficits that can persist after injury including altered proprioception, impaired motor control, muscle recruitment and heightened reliance on visual feedback. This suggests a shift from subconscious movement, to movements that require increased volitional control, which may contribute to increased risk of re-injury and thus impede return to sport.

Clinical question: Given the neurophysiological changes associated with anterior cruciate ligament (ACL) injury that persistent into the late stages of rehabilitation, does the integration of neurocognitive training into mid to late-stage rehabilitation protocols improve functional outcomes and reduce the risk of re-injury following ACL reconstruction (ACLR) in athletes?

Purpose: The purpose of Part 2 of this clinical commentary is to offer strategies to implement neurocognitive training elements into the traditional ACLR rehabilitation (in weeks 9+) and review updated testing metrics that may better discern an athlete's readiness to return to competition. A comprehensive rehabilitation framework incorporating both physical and neurocognitive components is proposed, aiming to improve both long-term outcomes and return to sport testing, as well as diminishing re-injury risk.

Conclusion: Updates to the traditional rehabilitation approach post ACLR, that include increased emphasis on neuroplastic, cognitive, and visual-motor capabilities exist. These help prepare athletes for the unpredictable and chaotic nature of the sporting environment and may facilitate a more effective return to sport for athletes, potentially mitigating the risk of re-injury.

Level of evidence: 5.

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Nutritional interventions to attenuate quadriceps muscle deficits following anterior cruciate ligament injury and reconstruction. Sports Medicine, Ahead of Print: 1-28, 2025.

Abstract - Following anterior cruciate ligament (ACL) injury, quadriceps muscle atrophy persists despite rehabilitation, leading to loss of lower limb strength, osteoarthritis, poor knee joint health and reduced quality of life. However, the molecular mechanisms responsible for these deficits in hypertrophic adaptations within the quadriceps muscle following ACL injury and reconstruction are poorly understood. While resistance exercise training stimulates skeletal muscle hypertrophy, attenuation of these hypertrophic pathways can hinder rehabilitation following ACL injury and reconstruction, and ultimately lead to skeletal muscle atrophy that persists beyond ACL reconstruction, similar to disuse atrophy. Numerous studies have documented beneficial roles of nutritional support, including nutritional supplementation, in maintaining and/or increasing muscle mass. There are three main mechanisms by which nutritional supplementation may attenuate muscle atrophy and promote hypertrophy: (1) by directly affecting muscle protein synthetic machinery; (2) indirectly increasing an individual's ability to work harder; and/or (3) directly affecting satellite cell proliferation and differentiation. We propose that nutritional support may enhance rehabilitative responses to exercise training and positively impact molecular machinery underlying muscle hypertrophy. As one of the fastest growing knee injuries worldwide, a better understanding of the potential mechanisms involved in quadriceps muscle deficits following ACL injury and reconstruction, and potential benefits of nutritional support, are required to help restore quadriceps muscle mass and/or strength. This review discusses our current understanding of the molecular mechanisms involved in muscle hypertrophy and disuse atrophy, and how nutritional supplements may leverage these pathways to maximize recovery from ACL injury and reconstruction.

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Quantifying muscle volume deficits among 38 lower extremity muscles in collegiate football athletes after anterior cruciate ligament reconstruction. The American Journal of Sports Medicine, Ahead of Print: 1-8, 2025.

Background: Quadriceps dysfunction is ubiquitous after anterior cruciate ligament (ACL) reconstruction (ACLR). Addressing quadriceps dysfunction is crucial to improve function, reduce the reinjury risk, and maintain long-term knee health. While deficits specific to the quadriceps are well documented, less is known about the effect of an ACL injury on other lower extremity muscle groups.

Purpose/hypothesis: The purpose of this exploratory analysis was to quantify and rank lower extremity muscle volume deficits using magnetic resonance imaging in collegiate football athletes after ACLR. It was hypothesized that the quadriceps muscles would present with the greatest deficits and that compensatory hypertrophy of muscles at adjacent joints such as the hip and ankle would be observed.

Study design: Cross-sectional study; Level of evidence, 3.

Methods: This study is a secondary analysis from an ongoing multicenter prospective cohort study involving Division I collegiate football athletes. Athletes who underwent primary unilateral ACLR (1 [3%] allograft, 2 [7%] quadriceps tendon autograft, 22 [73%] bone-patellar tendon-bone autograft, 5 [17%] hamstring tendon autograft) and magnetic resonance imaging were included. Muscle volumes (mL·kg-1·m-1) were quantified bilaterally from 38 lower extremity muscles using machine learning technology. Paired-samples t tests were performed between limbs for each muscle, which were then ranked and visualized in a forest plot based on standardized mean differences (surgical - nonsurgical limb).

Results: A total of 30 athletes (mean time from surgery, 27.9 ± 19.0 months) were included. The largest muscle volume deficits in the surgical limb were seen in the 3 uniarticular quadriceps muscles, followed by the biarticular triceps surae muscles. The rectus femoris and soleus did not show significant differences between limbs. Conversely, the fibularis muscle group had a greater muscle volume in the surgical limb compared with the nonsurgical limb. Most other muscle groups did not present significant differences between limbs.

Conclusion: Persistent quadriceps atrophy in a cohort of high-level athletes over 2 years after ACLR was highlighted in this study. Deficits in the gastrocnemius muscles, but not in the soleus, were also

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identified. This comprehensive approach examining various lower extremity muscles revealed latent muscle volume deficits present after ACLR.

Recent trends in quadriceps tendon and patellar tendon injuries in the National Football League. The Orthopaedic Journal of Sports Medicine, 13(2): 1-9, 2025.

Background: While quadriceps tendon (QT) and patellar tendon (PT) tears are relatively rare in the National Football League (NFL), the impact they can have on a player's season and career is significant. Despite the importance of extensor tendon injuries within the NFL, there is a lack of recent literature describing their risk factors, incidence, and impact on performance.

Purposes: To determine the incidence and impact of QT and PT injuries in the NFL between the 2009-2010 and 2022-2023 seasons and to explore the impact that player and injury characteristics have on injury risk, return to play (RTP), and player performance.

Study design: Descriptive epidemiology study.

Methods: Publicly available data were reviewed to capture all PT and QT tears reported in the NFL between the 2009-2010 and 2022-2023 seasons. RTP and performance metrics were recorded for each player during the season before injury and the first 2 seasons postinjury. Data were analyzed to determine statistically significant differences in proportions using chi-square tests, Fisher exact tests, or McNemar tests. Statistical significance was set at $P < .05$.

Results: A total of 80 extensor tendon tears (24 QT tears, 56 PT tears) were identified between the 2009-2010 and 2022-2023 NFL seasons. Risk factors for QT injury included body mass index ≥31, age ≥26 years, and >4 years of NFL experience. The risk for QT injury was higher in offensive and defensive linemen compared with skill players. The risk for PT injury was higher for offensive linemen compared with other position groups. Overall, 96.3% of extensor tendon injuries were season ending. The RTP rate for QT injuries was greater (58.3%) than that for PT injuries (55.4%). QT injuries resulted in a greater rate of return to prior performance level and a quicker return to performance compared with PT injuries.

Conclusion: The rates of QT and PT injury were higher in this study compared with those found in prior NFL studies. This study found similar RTP rates after QT injury but significantly lower RTP rates after PT

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injury compared with the previous literature. This highlights the significant, devastating, and increasing impact that extensor tendon injuries have on NFL players.

Relationships between the pain intensity at the time of the hamstring muscle injury and the hyperintense T2 weighted volume of the hamstring muscle lesion measured by MRI. Clinical Journal of Sport Medicine, Ahead of Print: 1-6, 2024.

Objectives: The primary objective of this study was to analyze the correlation between the pain experienced at the time of a hamstring muscle injury and the hyperintense T2 weighted volume of the lesion measured on MRI. The secondary objectives were to analyze the differences in this pain with the lesion grade and the hamstring muscle head involved.

Design: We performed a retrospective analysis of the data collected in a prospective, multicenter, noninterventional cohort study (HAMMER).

Patients: A total of 61 patients were included in this analysis.

Interventions: Patients with suspected hamstring muscle injury had an MRI after the injury.

Main outcomes measures: They were evaluated for their pain intensity experienced at the time of injury with a Numeric Pain Rating Scale (NPRS) from 0 to 10.

Results: There was a significant but weak correlation between the pain experienced at the time of hamstring muscle injury and the hyperintense T2 weighted volume ($r = 0.287$; $P = 0.025$). There were no significant differences in the pain according to the lesion grade ($F(3,57) = 0.982$; $P = 0.408$) nor according to the muscle involved ($F(3,57) = 1.013$; $P = 0.394$).

Conclusions: Pain at the time of injury seemed to be weakly correlated with the extent of muscle lesion at medical imaging. Pain at the time of injury alone seems not enough to give us a clinical meaningful indication of the MRI hyperintense T2 weighted volume.

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Return-to-play criteria following a hamstring injury in professional football (soccer): A scoping review. Research in Sports Medicine, Ahead of Print: 1-21, 2024.

Abstract - The present scoping review aims to describe the available criteria to determine Return-To-Play (RTP), propose methodological considerations and new research questions, and provide information to help practitioners in professional football make informed decisions regarding RTP following a hamstring strain injury (HSI) in professional male football. The following electronic databases were searched: PubMed, MEDLINE, web of science and SPORTDiscus using keywords related to HSI in elite football. All types of studies in English reporting at least one RTP criterion for professional football players who sustained an HSI were included. In total, 19 studies met the inclusion criteria. RTP criteria were divided into three categories: clinical, strength and performance criteria. Clinical criteria were also divided into sub-categories: absence of pain, hamstring flexibility, medical staff clearance, psychological readiness, surgeon's opinion and imaging. Practitioners working in professional male football could benefit from using a combination of criteria in their RTP battery of tests.

Sleep interventions in elite sport – A systematic review. South African Journal of Sports Medicine, 37: 1-16, 2025.

Background: Elite athletes encounter various situations and conditions that may disrupt their sleep, a crucial factor for optimal performance and well-being.

Objectives: The aim of this study was to synthesize existing research on the effect of sleep interventions on sleep quantity and quality in elite sports and to provide evidence-based guidance for athletes, coaches, and other stakeholders in elite sports who seek to enhance sleep quantity and quality.

Methods: This review followed the PRISMA guidelines, whereas the search was executed in September 2023 utilizing the electronic databases SCOPUS, PubMed and Web of Science. Studies were included if they met the inclusion criteria.

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Results: A total of 1014 studies were retrieved from the databases, and data extraction was performed on 32 studies. The included studies evaluated sleep hygiene education/strategies, acute cold exposure, light therapies, supplementation, neurostimulation/neurofeedback, and other (mindfulness and massage therapy) strategies. Sleep hygiene education was the most effective intervention to improve sleep quantity. Supplementation and light therapy interventions showed improved sleep quality and quantity. Additionally, cold water immersion and mindfulness showed improved sleep quality, but further studies are required for confirmation.

Conclusion: Future research should use reliable and valid methods to improve the quality of evidence and ensure conclusive findings.

The birth of modern sports nutrition: Tracing the path from muscle biopsies to creatine supplementation – A narrative review. Journal of the International Society of Sports Nutrition, 22: Supl2, 2463373, 2025.

Background: Modern sports nutrition has evolved through discoveries in muscle metabolism and dietary supplementation. Advances in muscle biopsy techniques revealed how diet influences muscle energetics and exercise performance. The establishment of the Metabolic Research Laboratory provided a platform for further investigation, leading to the identification of creatine monohydrate (CrM) as an effective ergogenic aid. This review outlines the historical development of sports nutrition research from the 1960s to the early 1990s, highlighting key breakthroughs in muscle glycogen metabolism, dietary interventions, and creatine supplementation.

Methods: We conducted a narrative review that combined personal accounts with seminal research studies. This approach allowed us to examine the contributions of Drs. Jonas Bergström and Eric Hultman-founders of the Metabolic Research Laboratory-as well as the early work of their postdoctoral colleague, Dr. Roger Harris.

Results: Muscle biopsy techniques enabled direct analysis of muscle metabolism, leading to insights into glycogen depletion and recovery. The Metabolic Research Laboratory advanced our understanding of muscle energetics and informed dietary strategies for enhancing performance. In 1992, the rediscovery

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of CrM supplementation demonstrated its capacity to increase intramuscular creatine levels, significantly improving exercise performance and recovery. These breakthroughs reshaped sports nutrition and expanded its relevance to clinical and aging populations.

Conclusion: The progression from early muscle metabolism research to the validation of CrM supplementation underscores how foundational laboratory discoveries have shaped modern sports nutrition. The work of the Metabolic Research Laboratory and its key investigators continues to inform applications in both performance enhancement and clinical health.

The reliability, bias, differences, and agreement between velocity measurement devices during the hang clean pull. Journal of Strength and Conditioning Research, 39(3): 289-204, 2024.

Abstract - This study examined the reliability, bias, differences, and agreement between the GymAware PowerTool (GA) and Tendo Power Analyzer (TENDO) during the hang clean pull (HCP). Fourteen resistance-trained men performed HCP repetitions with 20, 40, 60, 80, 100, 120, and 140% of their 1 repetition maximum hang power clean (1RM HPC) with GA and TENDO devices attached to the barbell. Least-products regression was used to examine instances of proportional and fixed bias for mean barbell velocity (MBV) and peak barbell velocity (PBV) between devices. In addition, Hedge's g effect sizes were calculated to determine the magnitude of the differences between devices. Excellent reliability was displayed by the GA for all measurements. While excellent reliability for the TENDO was displayed for MBV across all loads, only moderate-good reliability was present for PBV at loads >80% 1RM. The TENDO displayed proportional bias for both MBV and PBV as well as fixed bias for PBV at 140% 1RM compared with the GA. Despite the small effect sizes that existed between devices for both MBV (g = 0.35-0.57) and PBV (g = 0.23-0.54), none of these differences were practically meaningful. The GA and TENDO are reliable devices that can measure MBV and PBV accurately during the HCP; however, practitioners should note that the TENDO may overestimate MBV and PBV to a small extent (3.7-6.8% and 1.8-2.9%).

Practical Applications - The GA and TENDO can provide reliable MBV and PBV measurements during the HCP performed using loads ranging 20–140% 1RM HPC. It should be noted, however, that the TENDO

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may become less reliable as loads increase, possibly due to increases in horizontal displacement. While the TENDO may overestimate barbell velocity, this may be accounted for if the device is consistently used within training. Despite the small differences in MBV and PBV between the GA and TENDO shown across all loads, strength and conditioning practitioners and sport scientists should note that VBT devices should not be used interchangeably. Moreover, users should be aware of the limitations of the factory settings of the TENDO as well as its inability to account for horizontal velocity. Researchers interested in using VBT devices for athlete monitoring purposes should continue to evaluate the reliability of different devices with the HCP and other weightlifting derivatives.

Use of phase angle as angle as an indicator of overtraining in sport and physical training. Journal of Translational Medicine, 22: 1084, 2024.

Abstract - The use of bioelectrical impedance analysis (BIA) is now well established in healthcare as an essential support tool for patient management in various clinical settings. Its use in sports is rapidly expanding due to the valuable insights it offers, helping to better structure athletes' diets and training programs, thereby optimizing their performance. In the context of sport, however, there is a consensus regarding the importance of proper interpretation of BIA-derived data, which cannot be limited to mere estimation of body composition. In this sense, therefore, the evaluation and interpretation of raw bioelectrical parameters, including resistance, reactance, and phase angle (PhA) is of relevant importance. The assessment of PhA is particularly significant in the context of sports, as it is closely linked to key factors such as muscle mass, strength, and overall muscle quality. However, the existing relationship between PhA and systemic, and loco-regional inflammation, which, in a broader sense, is the rationale behind its use for assessing and monitoring localized muscle damage. Thus, the importance of PhA monitoring during training becomes evident, as it plays a crucial role in assessing and potentially identifying functional impairments, such as overtraining syndrome, as well as muscle injury and related changes in fluid distribution, at an early stage. The aim of this review is to provide the scientific basis necessary to consider the use of whole-body PhA as an indicator of overtraining.

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